Experiment No: 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Blockchain is a decentralized and distributed digital ledger that records transactions securely across multiple systems. It ensures transparency, security, and immutability, meaning once data is recorded, it cannot be changed.

Ethereum is a popular blockchain platform that allows the execution of Smart Contracts. Smart contracts are self-executing programs written in Solidity that automatically perform actions when predefined conditions are met, eliminating the need for intermediaries.

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks. It stores user accounts and private keys securely and allows users to send, receive, and sign transactions. It also acts as a bridge between Remix IDE and the blockchain network.

Testnets such as Sepolia and RSK Testnet are used for testing purposes. They provide free tokens, allowing developers to deploy and test smart contracts without using real cryptocurrency, reducing financial risk.

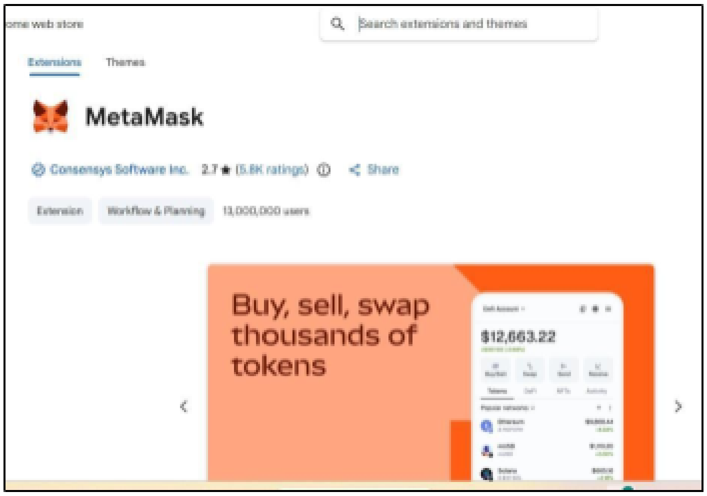
Remix IDE is a web-based development environment used to write, compile, deploy, and test smart contracts. It provides a user-friendly interface and supports direct integration with MetaMask for deploying contracts on blockchain networks.

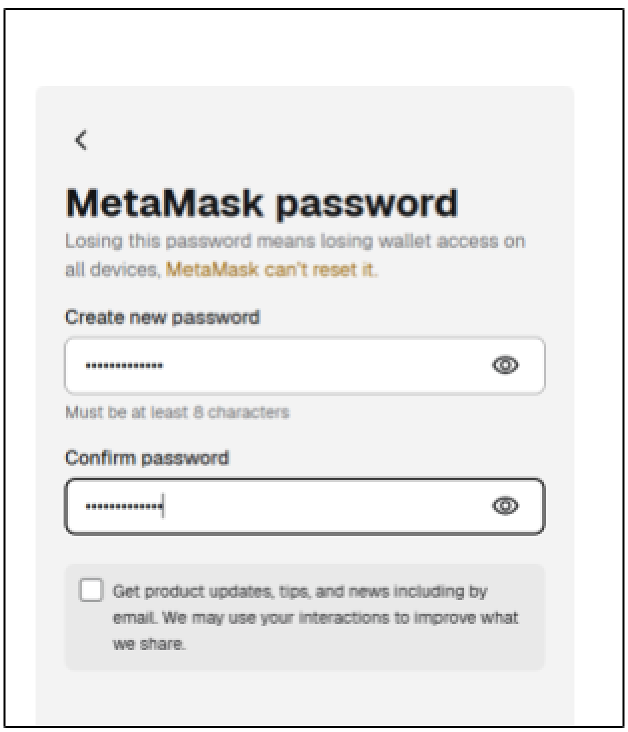
The smart contract deployment process includes writing the contract in Solidity, compiling it to remove errors, connecting MetaMask to Remix, and deploying the contract on a testnet by paying gas fees using test tokens.

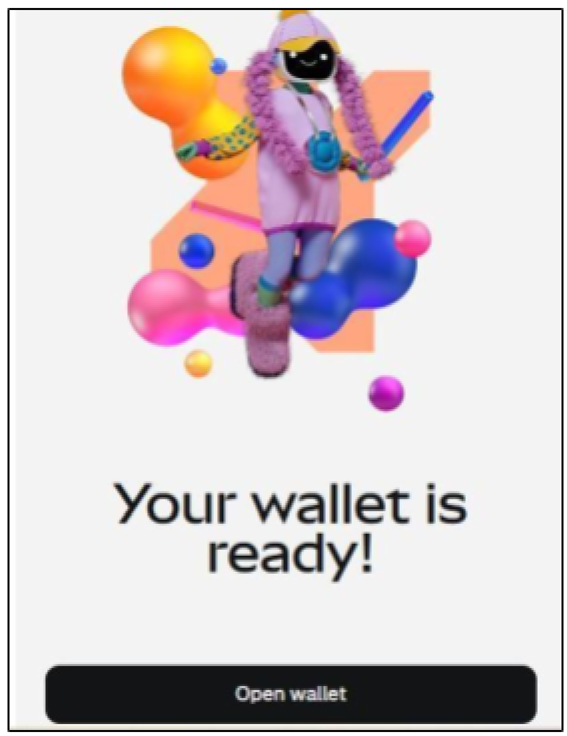
A blockchain transaction refers to any interaction with the network, such as deploying a contract or calling its functions. Each transaction requires authentication through a digital signature and is permanently recorded on the blockchain. These transactions can be verified using blockchain explorers.

Implementation

Step 1: Install MetaMask
Install MetaMask extension and create a wallet

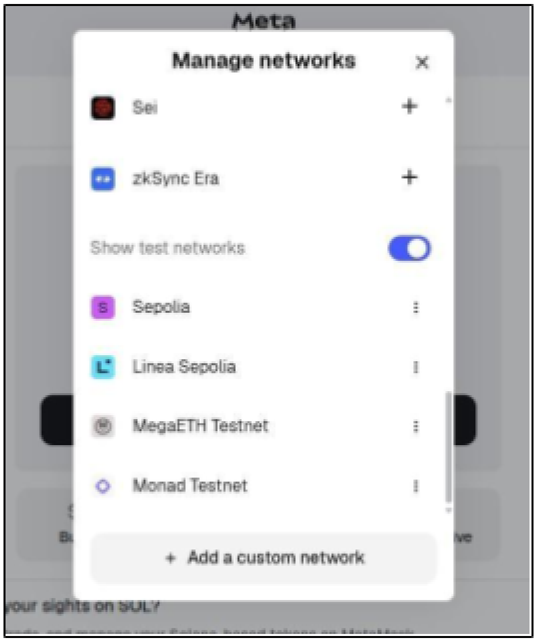


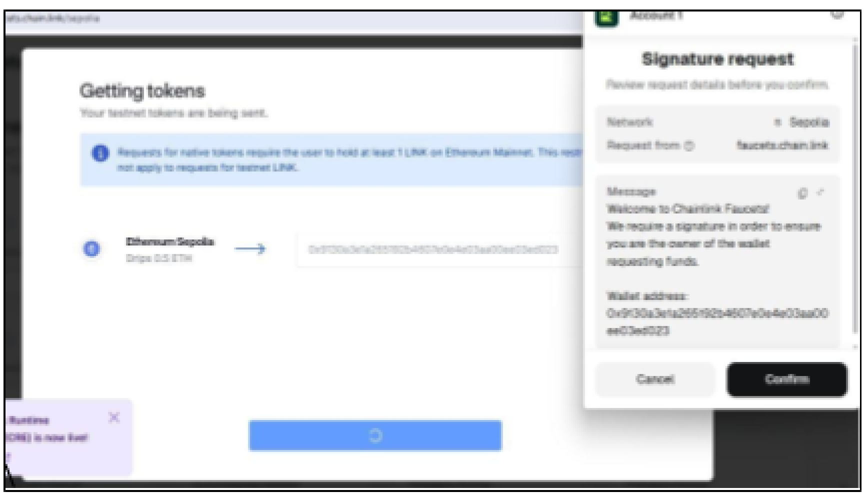


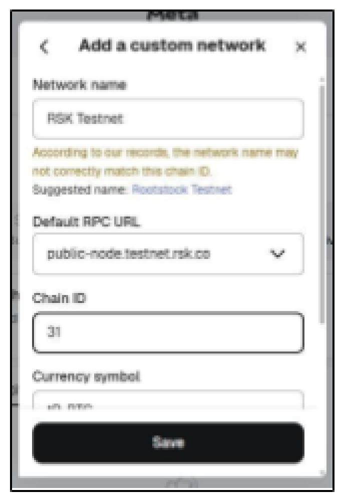


Step 2: Add Sepolia Testnet

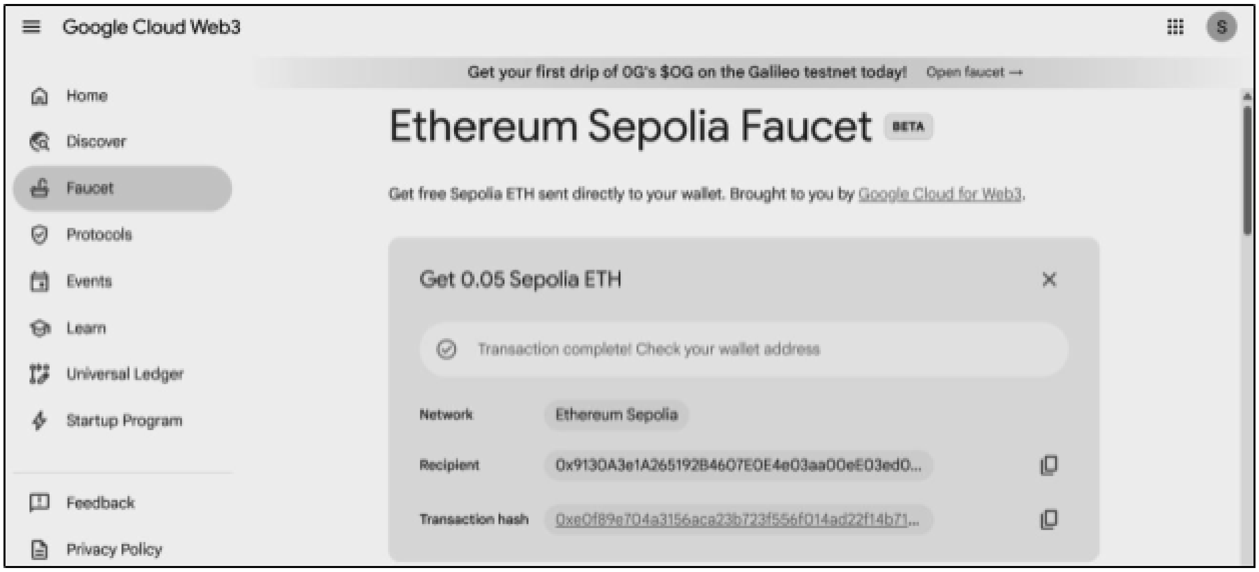
Configure MetaMask to connect to Sepolia network.

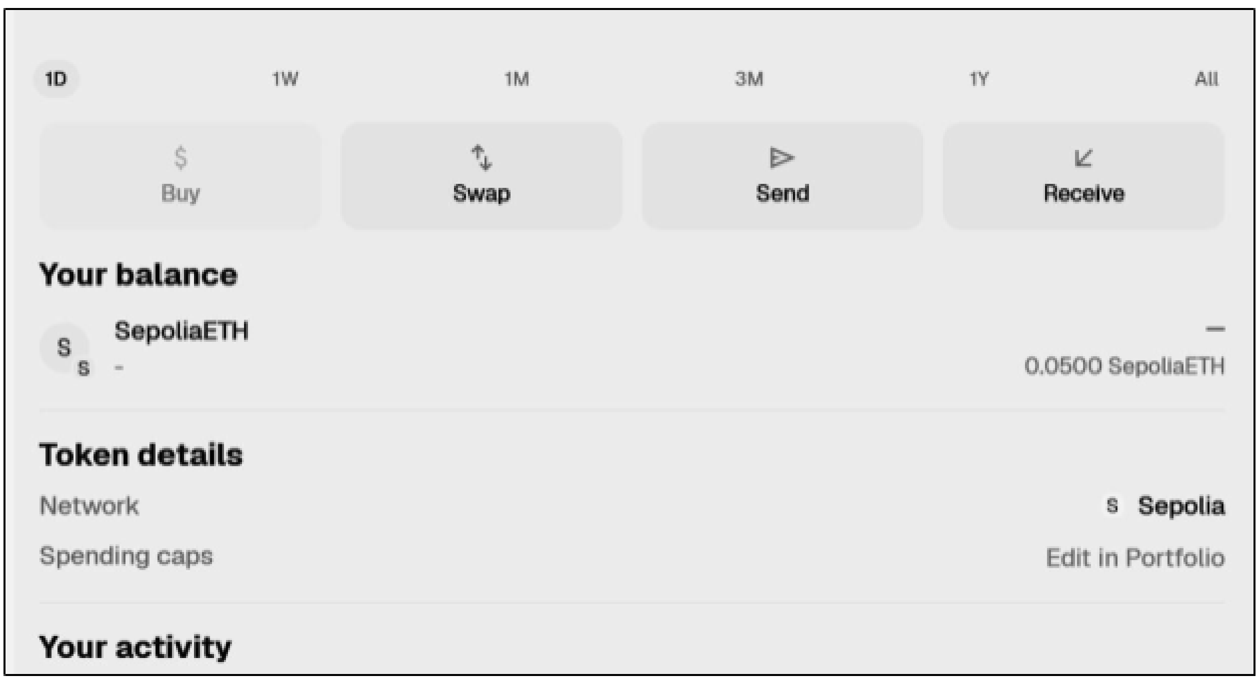




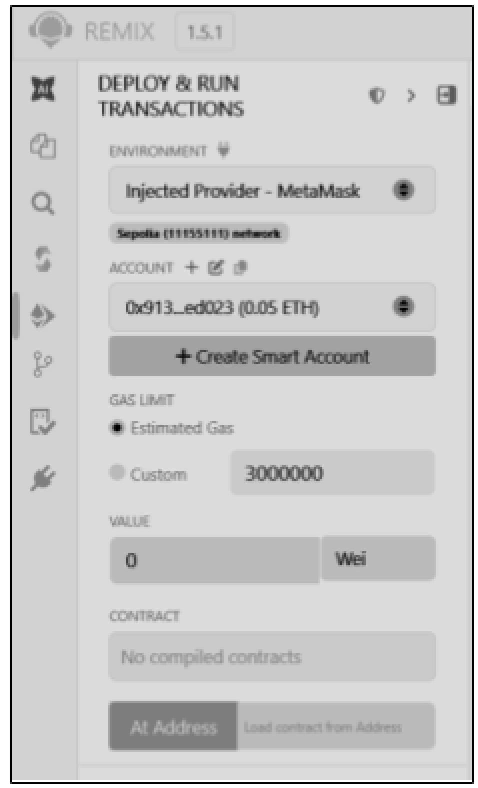


Step 3: Fund Wallet
Use faucet to get test ETH



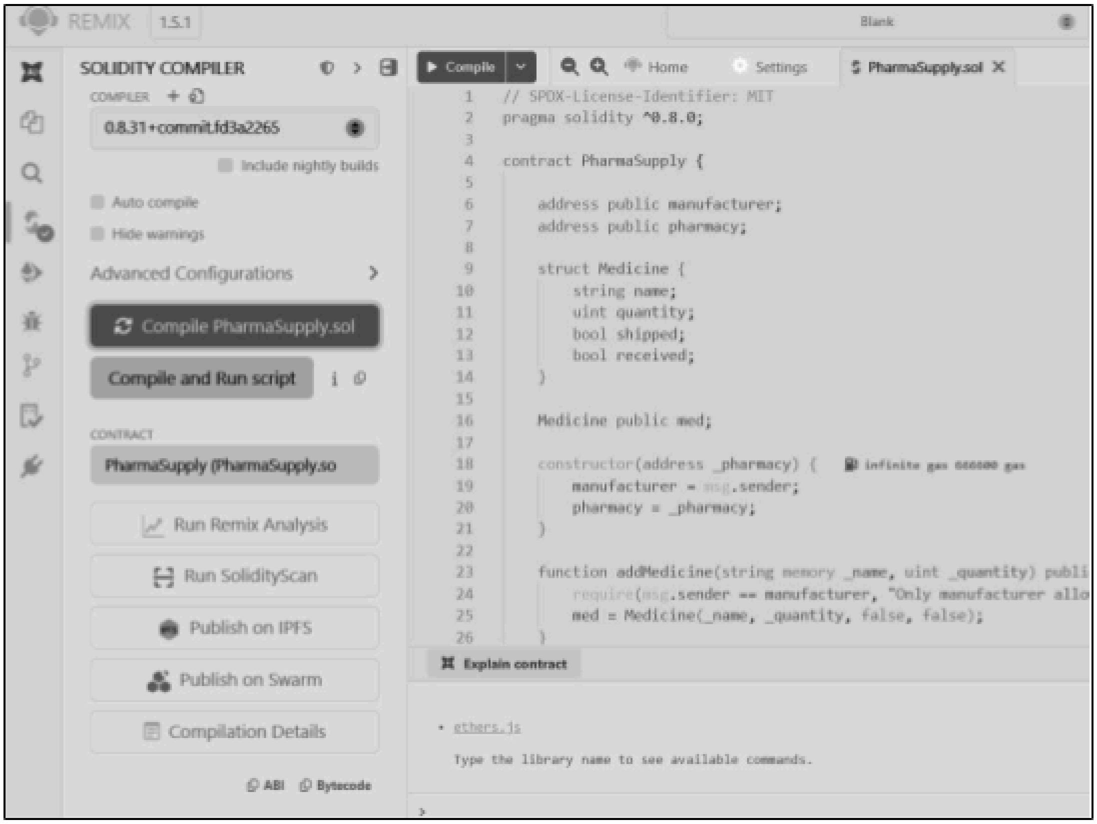


Step4: Connect the Sepolia Testnet / RSK Testnet to Remix IDE
1. In the Remix IDE, select the Environment as Injected Web3.
Note : Injected Web3 connects Remix with the active account in Metamask.

Compile the Contract

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile PharmaSupply.sol

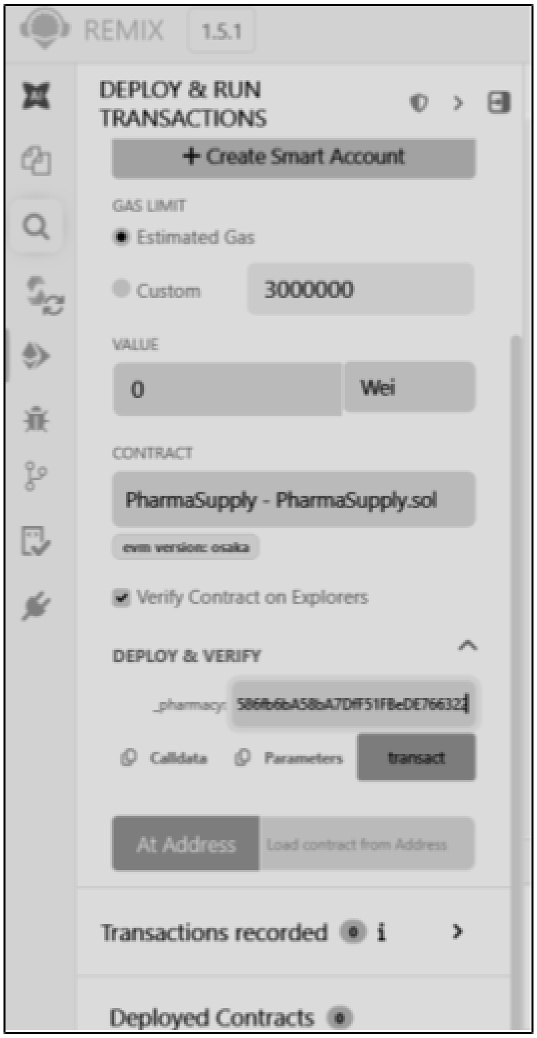


Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click

Connect Make sure:

- Network = Sepolia Test Network
- You have 0.05 Sepolia ETH



Deploy Contract

Your constructor needs a pharmacy address

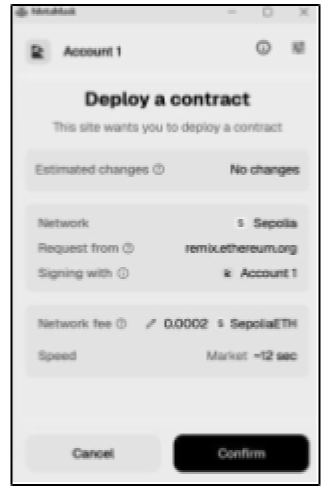
Steps:

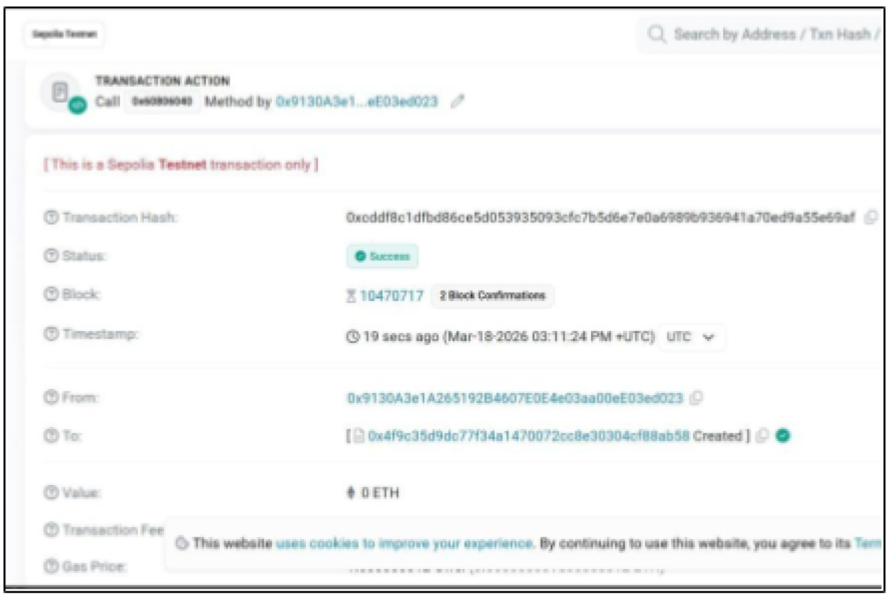
1. Copy your second MetaMask account address (pharmacy)
    - MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction

Wait a few seconds…

After success → your contract appears under Deployed Contracts


Accounts
Search your accounts
Account 1
$0.00
Pharmacy
$0.00
Add account
Add wallet


Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Signing with
Account 1
Network fee
0.0002
SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


Search by Address / Txn Hash /
TRANSACTION ACTION
Call
Method by 0x9130A3e1...eE03ed023
[This is a Sepolia Testnet transaction only]
Transaction Hash:
0xcddf8c1dfbd86ce5d053935093cfc7b5d6e7e0a6989b936941a70ed9a55e69af
Status:
Success
Block:
10470717
2 Block Confirmations
Timestamp:
19 secs ago (Mar-18-2026 03:11:24 PM +UTC)
UTC
From:
0x9130A3e1A265192B4607E0E4e03aa00eE03ed023
To:
[0x4f9c35d9dc77f34a1470072cc8e30304cf88ab58 Created]
Value:
0 ETH
Transaction Fee
Gas Price:
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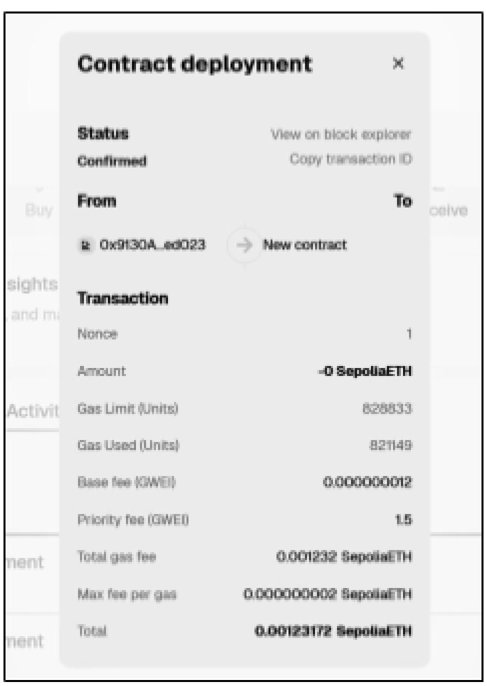
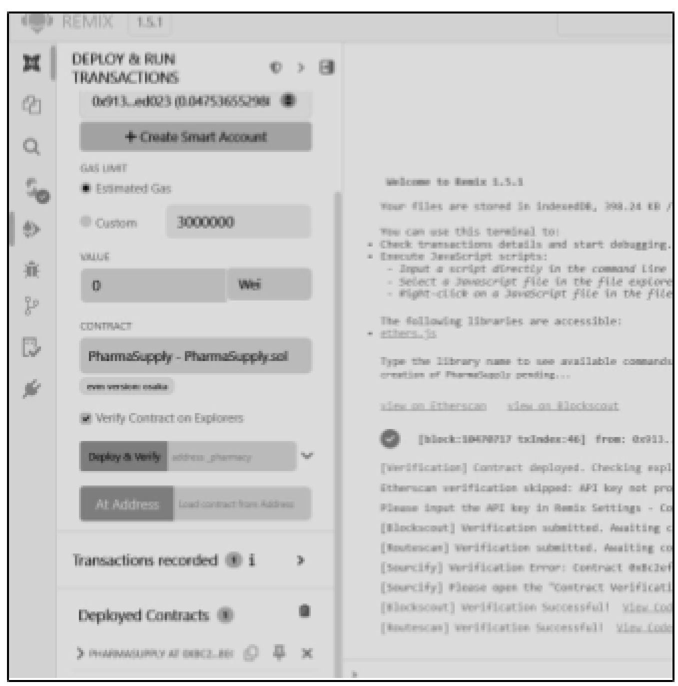

Contract deployment
Status
View on block explorer
Confirmed
Copy transaction ID
From
To
0x9130A...ed023
New contract
Transaction
Nonce
1
Amount
-0 SepoliaETH
Gas Limit (Units)
828833
Gas Used (Units)
821149
Base fee (GWEI)
0.000000012
Priority fee (GWEI)
1.5
Total gas fee
0.001232 SepoliaETH
Max fee per gas
0.000000002 SepoliaETH
Total
0.00123172 SepoliaETH

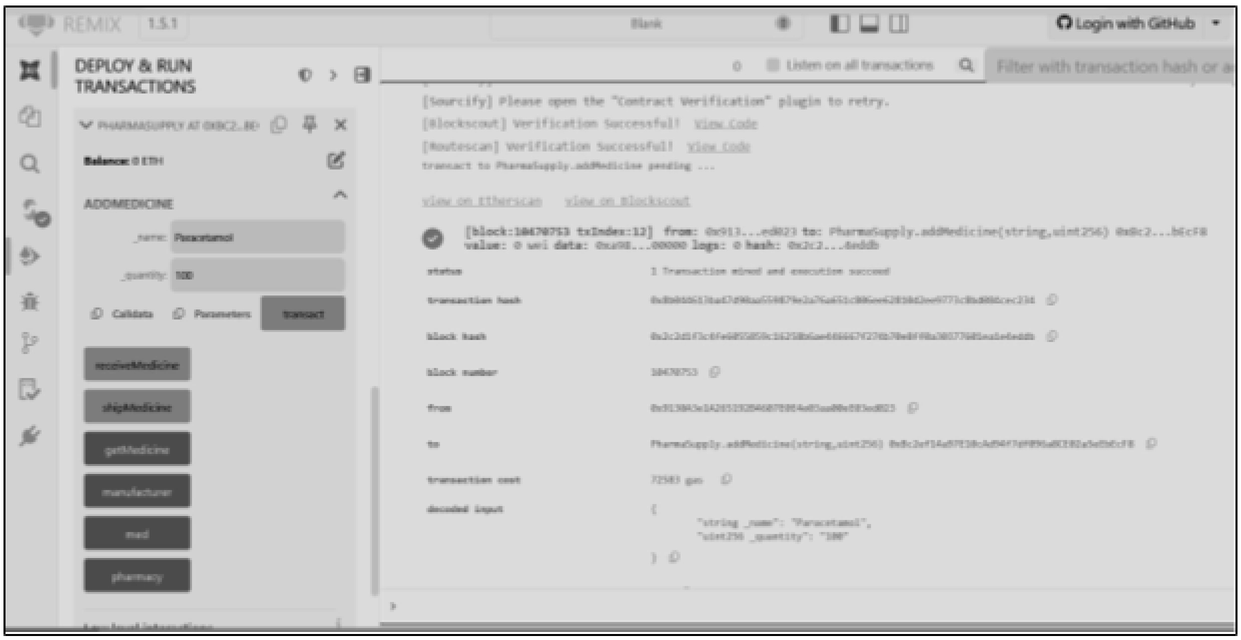

REMIX
DEPLOY & RUN TRANSACTIONS
Create Smart Account
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
PharmaSupply - PharmaSupply.sol
Verify Contract on Explorers
Deploy & Verify
At Address
Transactions recorded
Deployed Contracts
Welcome to Remix 1.5.1

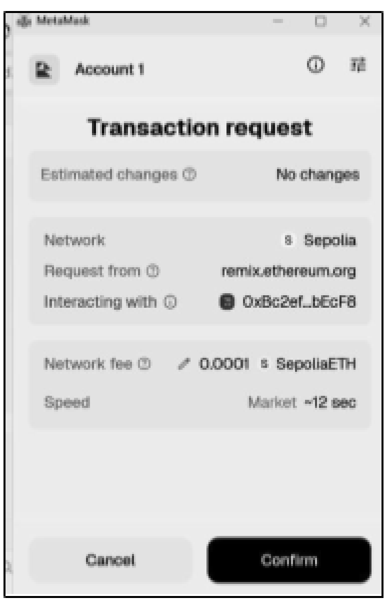
Interact with Contract

Step 1: Add Medicine (Manufacturer only)

- Make sure MetaMask is on Account 1 (manufacturer)
- In Remix:
  - Enter:
    - _name: "Paracetamol"
    - _quantity: 100
- Click addMedicine

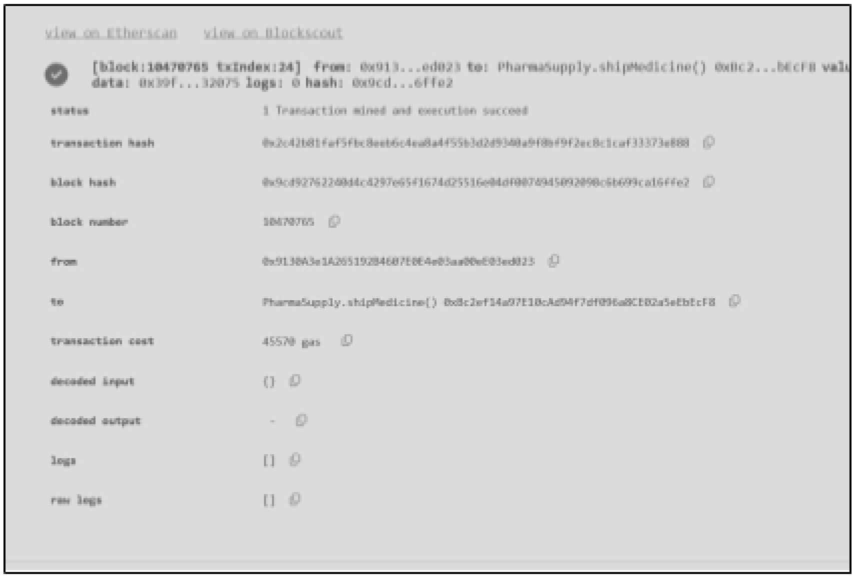
Confirm in MetaMask

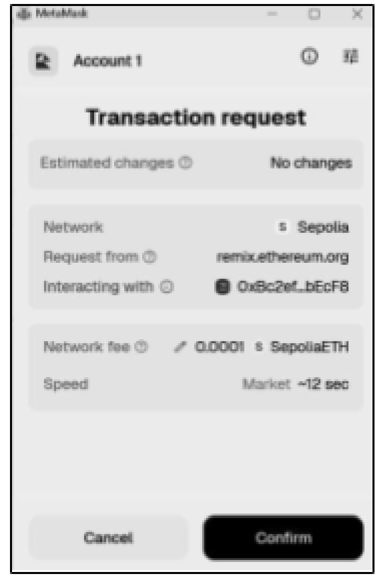


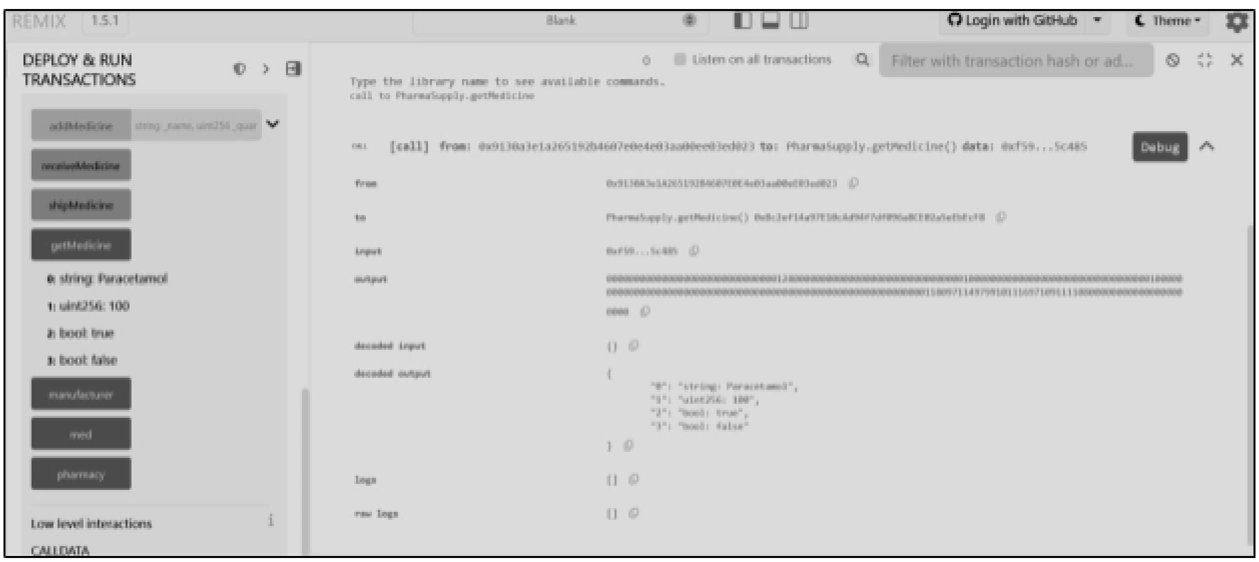


Step 2: Ship Medicine

- Still on Account 1
- Click shipMedicine
- Confirm transaction

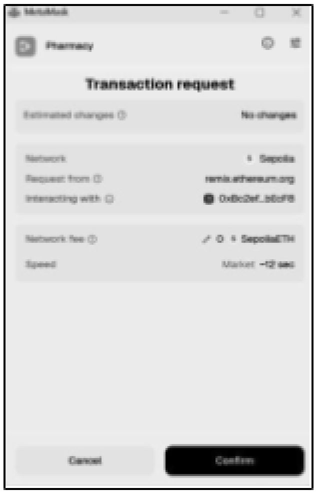






Step 3: Receive Medicine (Pharmacy)

- Switch MetaMask to Account 2 (pharmacy)
- Click receiveMedicine
- Confirm transaction



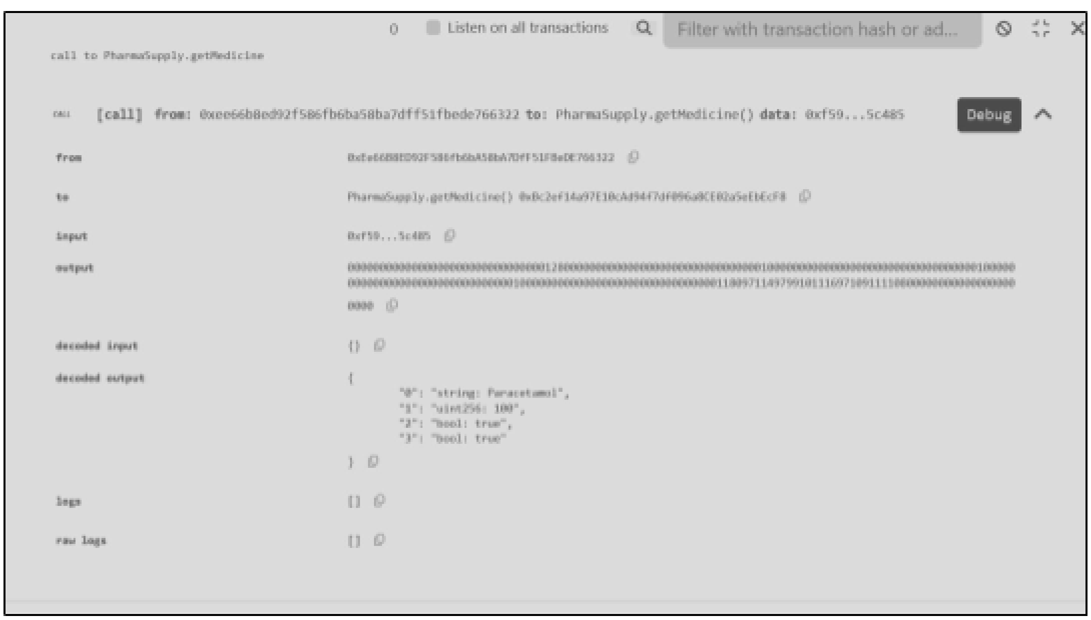
Step 4: Check Data

- Click

getMedicine Output will

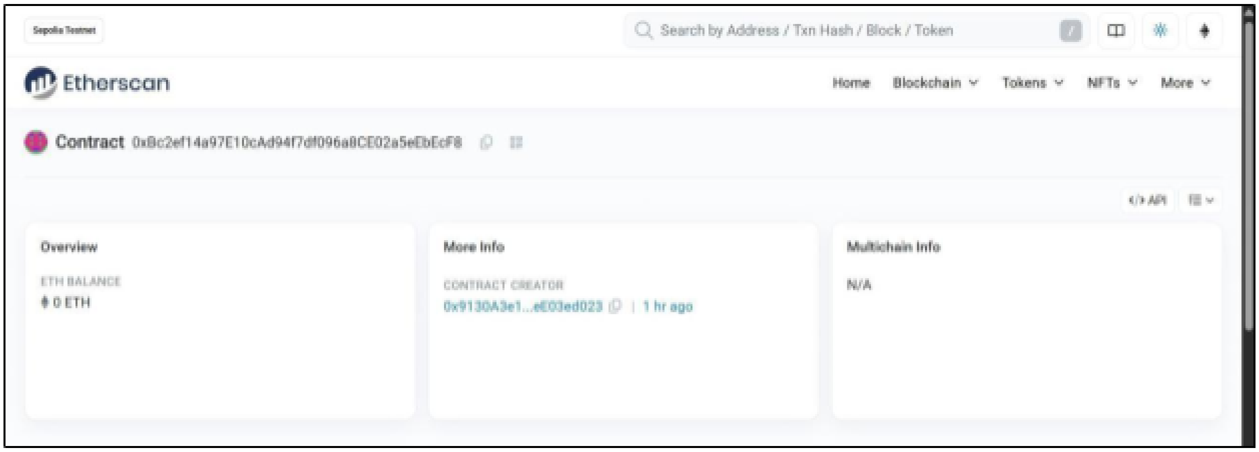
be like: "Paracetamol"

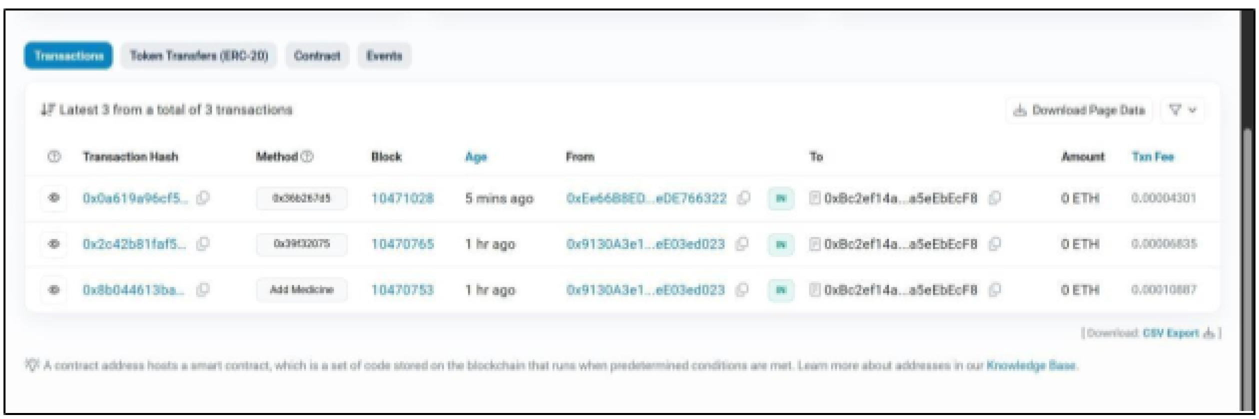
100
true (shipped)
true (received)



Compiled PharmaSupply.sol

Copy contract address → paste in Etherscan (Sepolia) See all transactions + status





Conclusion

The experiment successfully demonstrated MetaMask integration with Remix IDE and execution of blockchain transactions. It provided hands-on experience with smart contracts, ensuring secure and transparent interactions on a testnet